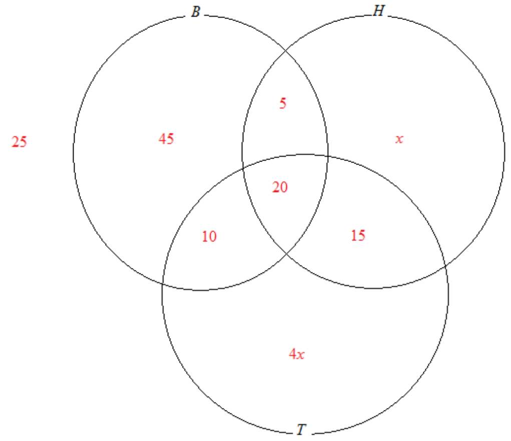
Question		Scheme		Mark	Notes	
1	(a)	$348\times\frac{100}{60}$ oe	(\$) 580	2	M1 A1	
	(b)	$"\$580"\times\frac{75}{100}-\$348$ OR $(0.40-0.25)\times"580"$ OR $(0.75-0.60)\times"580"$	(\$)87	2	M1 A1	
2			$x=-14$ $y=10$	4	M1	Rearranging st coef of x or y is the same in both eqns OR isolating x or y
					M1	(DEP) Substituting expression (or value correctly obtained) for x or y to obtain y or x **NB:** Allow a total of 1 slip in both M marks.
					A1 A1	

Question		Scheme		Mark	Notes
3	(a)	$576 = \dfrac{\alpha}{\left(\frac{1}{2}\right)^3}$ $\alpha = 72$ $\quad\therefore f = \dfrac{72}{r^3}$		3	M1 A1 A1
	(b)	$f = 5 + \dfrac{1}{t} = \dfrac{"72"}{2^3}$ $\quad(= 9)\quad$ (oe)	$t = \dfrac{1}{4}$	2	M1 A1

Question	Scheme		Mark	Notes
4	One of $(1,1)$: $-7+2x^2=1$ (ie 1st column) $(2,1)$: $-21-4x^2=-37$ $(3,1)$: $35-6x^2=11$	$x=2$	6	M1 A1
	One of $(1,2)$: $1+2(\text{"}x\text{"}+2y)=1$ (ie 2nd column) $(2,2)$: $3-4(\text{"}x\text{"}+2y)=3$ $(3,2)$: $-5-6(\text{"}x\text{"}+2y)=-5$	$y=-1$		M1 (DEP) A1
	One of $(1,3)$: $-\text{"}x\text{"}z-2\text{"}y\text{"}=-4$ (ie 3rd column) $(2,3)$: $-3\text{"}x\text{"}z+4\text{"}y\text{"}=-22$ $(3,3)$: $5\text{"}x\text{"}z+6\text{"}y\text{"}=24$	$z=3$		M1 (DEP) A1

Question		Scheme	Mark	Notes
5	(a)		4	B1 25 correctly positioned B1 5, 10 and 15 correctly positioned B1 45 and 20 correctly positioned B1 4x correctly positioned in T and x correctly positioned in H
	(b)	150 = 25 + “45” + ”5” + x + “10” + 20 + “15” + 4x (oe) (ie 150 = their 8 values)	1	B1 ft
	(c)	(eg 150 = “120” + 5x (oe)) (cao) $x = 6$	2	M1 Collecting “their” two x terms and equating them to “their” 7 constant values A1
	(d)	$\left(\frac{"10"+"20"}{"45"+"5"+"10"+"20"}=\right)$ $\frac{"30"}{"80"}$ (oe), "0.375", "37.5"%	1	B1 Ft **NB**: ft on their diagram

Question		Scheme	Mark	Notes
6	(a)	$\frac{3}{5}$, 0.6	1	B1
	(b)	$\frac{3}{2}$ OR 1.5 OR not 3/2	1	B1
	(c)	$y(2x-3)=6$ (oe) OR $x(2y-3)=6$ (oe) $\mathrm{h}^{-1}: x \mapsto \frac{6+3x}{2x}$, $\frac{3(2+x)}{2x}$, $\frac{3}{x}+\frac{3}{2}$, $\mathrm{h}^{-1}=\frac{6+3x}{2x}$ (oe)	2	M1 A1
	(d)	$18x-x(2x-3)=3(2x-3)$ (removing denominators, oe, allow 1 minor slip) $2x^2-15x-9(=0)$ (oe) $x=\frac{-(-15)\pm\sqrt{\left((-15)^2-4\times2\times(-9)\right)}}{2\times2}$ **NB:** on their trinomial quadratic. –0.558 8.06		M1 A1 M1 (INDEP) A1 A1

Question		Scheme		Mark	Notes
7	(a)	$65 < t <\leq 70$ fd = 4 (8 x 1cm squares) units		5	B1
		$70 < t \leq 80$ freq = 50 runners			B1
		$80 < t \leq 95$ fd=4units			B1
		$95 < t \leq 115$ fd = 4.5 units			B1
		$115 < t \leq 140$ freq = 75 and fd = 3 units			B1 ft
	(b)		$95 < t \leq 115$	1	B1 Ft **NB:** ft on “50” for $70 < t \leq 80$
	(c)	Using a correct mid-pt	98 (minutes)	4	M1
		At least 3 correct products			M1 (DEP)
		$\dfrac{10\times 62.5+20\times 67.5+"50"\times 75+60\times 87.5+90\times 105+"75"\times 127.5}{305}$			M1 (DEP)
		$\left(=\dfrac{625+1350+"3750"+5250+9450+9562.5}{305}=\dfrac{29987.5}{305}\right)$			A1 (cao)

Question	Scheme	Mark	Notes
8 (a) (i)	$\overrightarrow{AB} = 8\mathbf{b} - 4\mathbf{a}$	1	B1
(ii)	$\overrightarrow{PO} = -\mathbf{a}$	1	B1
(b)	$\overrightarrow{PQ} = \alpha"(8\mathbf{b} - 4\mathbf{a})" = -\mathbf{a} + \frac{8}{m}\mathbf{b} \quad \left(= \overrightarrow{PO} + \overrightarrow{OQ}\right)$ $\alpha = \frac{1}{4}$ $m = 4$	3	M1 A1 A1 **NB:** Cand. must use vectors as required by question.
(c)	$\overrightarrow{PR} = \overrightarrow{PA} + \overrightarrow{AR} = 3\mathbf{a} + \frac{1}{n}"(8\mathbf{b} - 4\mathbf{a})"$ $\overrightarrow{PR} = \left(3 - \frac{4}{n}\right)\mathbf{a} + \frac{8}{n}\mathbf{b}$, $\quad 3\mathbf{a} - \frac{4}{n}\mathbf{a} + \frac{8}{n}\mathbf{b}$ $\quad , \frac{3n\mathbf{a} - 4\mathbf{a} + 8\mathbf{b}}{n}$	2	M1 A1 **NB:** So **a** and **b** terms separated
(d)	*PR* parallel to *OB* means "comp of **a**" in $\overrightarrow{PR}$ above is zero (**OR** since triangles *AOB* and *ARB* are similar, $\frac{AP}{AO} = \frac{3}{4} = \frac{PR}{OB}$, Comp of **b** in (c) means that $\therefore \overrightarrow{PR} = 6\mathbf{b} = \frac{8}{n}\mathbf{b}$ (M1)**)** $n = \frac{4}{3}$	2	M1 A1
(e)	Triangles *OAB* and *OPQ* are similar (oe) $\therefore \ \lvert\Delta OAB\rvert = 4^2 \times \lvert\Delta OPQ\rvert$ *APQB* = 150 = Triangle *OAB* □□Triangle *OPQ* $\therefore 150 = 4^2\lvert\Delta OPQ\rvert - \lvert\Delta OPQ\rvert$ (oe) $\therefore \lvert\Delta OPQ\rvert = \mathbf{10}\left(\text{cm}^2\right)$	3	M1 M1 (DEP) A1

Question		Scheme	Mark	Notes
9	(a)	Triangle *S* drawn and labelled	1	B1
	(b)	Triangle *T* drawn and labelled $\left(\Delta T = \begin{pmatrix} 2 & 3 & 3 \\ 4 & 4 & 6 \end{pmatrix}\right)$	2	B2 (-1ee)
	(c)	Either point (-2,2) indicated **OR** At least two construction lines through (-2,2) Triangle *U* $\left(\Delta U = \begin{pmatrix} -6 & -7 & -7 \\ 0 & 0 & -2 \end{pmatrix}\right)$ **NB:** Award M1 A2 if (-2,2) not indicated and no construction lines but Δ*U* drawn correctly Award M1 A1 A0 if Δ*U* drawn correctly except for one Vertice.	3	M1 A2 (-1ee)
	(d)	Triangle *V* drawn and labelled $\left(\Delta V = \begin{pmatrix} -1 & -2 & -2 \\ -1 & -1 & -3 \end{pmatrix}\right)$ **NB:** ft on "triangle *U*"	2	B2) ft (-1ee
	(e)	$\begin{pmatrix} -3 & 1 \\ 1 & 1 \end{pmatrix}$"$\begin{pmatrix} -1 & -2 & -2 \\ -1 & -1 & -3 \end{pmatrix}$" Triangle *W* drawn and labelled $\left(\Delta W = \begin{pmatrix} 2 & 5 & 3 \\ -2 & -3 & -5 \end{pmatrix}\right)$		M1 A2 (-1ee)
	(f)	-4	1	B1
	(g)	1 : 4	1	B1

Question	Scheme	Mark	Notes
10 (a)	$\sin 25 = \frac{20}{AB}$	2	M1 A1
	47.3240→**47.3** (cm)		
(b)	$\cos 20 = \frac{FC}{15}$	2	M1 A1
	14.0954→**14.1** (cm)		
(c)	$AC^2 = "AB"^2 + 15^2 - 2\times "AB" \times 15 \times \cos 95$	3	M1
	$AC = \sqrt{\left("AB"^2 + 15^2\right) - \left(2\times "AB" \times 15 \times \cos 95\right)}$		M1 (DEP) A1
	50.9 (cm)		
(d)	Method 1: $ABCD = \lvert\Delta ABC\rvert + \lvert\Delta ACD\rvert$	6	M1 (DEP)
	Scheme: Δ*ABC*: M1 (angle for area formula), M1(area formula)		M1
	Δ*ACD*: M1 (angle or side for area formula), M1(area formula)		
	A*BCD*: M1 (adding areas) A1		
	$\angle ABC = 25 + (180 - 90 - 20) \quad (=95)$		
	NB: $\angle ABC$ must be evaluated to **95**		
	$\lvert\Delta ABC\rvert = \frac{1}{2}\times 15 \times "AB" \times \sin "\angle ABC" \qquad \left(= \begin{cases} 353.6 & \text{using 4sf} \\ 353.4 & \text{using 3sf} \end{cases}\right)$		M1 (DEP)) M1

	(Point *X* is st *AD* is perpendicular to *CX*			
	$\therefore AX = 20 + "FC"$)			(M1)
	$\therefore \cos\angle CAD = \dfrac{"AX"}{"AC"}$ $\left(\angle CAD = \begin{cases} 47.94^\circ \text{ using 3sf answers} \\ 47.92^\circ \text{ using 4sf answers} \end{cases}\right)$			(M1
	$\therefore \lvert \Delta ACD \rvert = \dfrac{1}{2} \times 40 \times "AC" \times \sin "\angle CAD"$ $\left(= \begin{cases} 755.2 & \text{using 4sf} \\ 755.8 & \text{using 3sf} \end{cases}\right)$			
	[OR $\therefore CX = \sqrt{"AC"^2 - "AX"^2}$			
	$\therefore \lvert \Delta ACD \rvert = \dfrac{1}{2} \times 40 \times "CX"$]			
	(OR $\angle ABC = 25 + (180 - 90 - 20)$ $(= 95)$			M1) (DEP)) (M1 (DEP)) (M1 (DEP)) (M1
	NB: $\angle ABC$ must be evaluated to **95**			
	$\angle BAC = \sin^{-1}\left(\dfrac{15 \times \sin 95}{"50.9"}\right)$ $(= 17.07)$			
	$\lvert \Delta ABC \rvert = \dfrac{1}{2} \times "47.324" \times "50.9" \times \sin "\angle BAC"$			

	$\angle CAD = 65 - "17.07" \quad (= 47.93)$) $\therefore \lvert \Delta ACD \rvert = \frac{1}{2} \times 40 \times "50.9" \times \sin"(65 - 17.07)"$ Finally: $\therefore ABCD = "\lvert \Delta ABC \rvert" + "\lvert \Delta ACD \rvert" \quad \left(= \begin{cases} 1108.8 & \text{using 4sf} \\ 1109.2 & \text{using 3sf} \end{cases} \right)$ *ABCD* = **1110** (cm^2)		M1 (DEP) A1
	Method 2: *ABCD* = ($\Delta ABE + \Delta BCF$) + *CFED* **Scheme: Δ*ABE* + Δ*BCF*:** M1(full method for area) ***CFED*:** M1(side or angle need to find *CX*), M1(full method for *CX*), M1(area formula for *CFED*) ***ABCD*:** M1(adding areas), A1 *ABCD* = ($\lvert \Delta ABE \rvert + \lvert \Delta BCF \rvert$) + *CFED* ($\Delta ABE + \Delta BCF$) = $\left(\frac{1}{2} \times "AB" \times 20 \times \sin 65 \right) + \left(\frac{1}{2} \times "FC" \times 15 \times \sin 20 \right)$ M1		M1 M1 M1 (DEP) M1 (DEP) M1 (DEP) A1

	$\left(=\begin{cases}464.852 & \text{using 3sf}\\ 465.06 & \text{using 4sf}\end{cases}\right)$ Point *X* is st *AD* is perpendicular to *CX* $\therefore AX = 20 + "FC"$ M1 $\therefore CX = \sqrt{"AC"^2 - "AX"^2}$ $\left(=\begin{cases}37.79 & \text{using 3sf}\\ 37.76 & \text{using 4sf}\end{cases}\right)$ M1 (DEP) (OR $\tan 25 = \frac{20}{BE}$ (*BE* = 42.89) (M1) *FE* = *CX* = $"BE" - 15\sin 20$ (M1(DEP))) $\therefore CFED = \frac{1}{2} \times "CX" \times ("FC" + 20)$ $\left(=\begin{cases}644.32 & \text{using 3sf}\\ 643.71 & \text{using 4sf}\end{cases}\right)$ M1 (DEP) ∴ *ABCD* = "($\Delta BCF + \Delta ABE$)" + "*CFED*" $\left(=\begin{cases}1108.8 & \text{using 4sf}\\ 1109.2 & \text{using 3sf}\end{cases}\right)$ M1 (DEP) *ABCD* = **1110** (cm^2)		
	Method 3: $\Delta ABC + \Delta ACX + \Delta CXD$ **Scheme: ΔABC**:M1 (angle for area formula), M1(area formula)	6	M1 (DEP) M1 M1 M1 (DEP)

	Δ*ACX*: M1(full method for area formula)		M1 A1
	Δ*CXD*: M1(full method for area formula)		
	ABCD: M1 (Adding areas) A1		
	ABCD = $\underline{\lvert\Delta ABC\rvert+\lvert\Delta ACX\rvert+\lvert\Delta CXD\rvert}$		
	$\angle ABC = 25+(180-90-20)\quad(=95)$ M1		
	NB: $\angle ABC$ must be evaluated to **95**		
	$\lvert\Delta ABC\rvert = \frac{1}{2}\times 15\times "AB"\times\sin"\angle ABC"\qquad \left(=\begin{cases}353.6 & \text{using 4sf}\\ 353.4 & \text{using 3sf}\end{cases}\right)$ M1(DEP)		
	(Point *X* is st *AD* is perpendicular to *CX*		
	$\therefore AX = 20+"FC"$)		
	$\left(BE = 20\tan 65 = 42.89 \quad \text{and} \quad BF = 15\sin 20 = 5.130 \quad \therefore\ FE = 37.7598\right)$		
	$\lvert\Delta ACX\rvert = \frac{1}{2}\times"34.095"\times"37.76"\quad(=643.718)$ M1		
	$\left(DX = 20-"14.095" = 5.905\right)$		
	$\lvert\Delta CXD\rvert = \frac{1}{2}\times 37.76\times 5.905\quad(=111.479)$ M1		

	∴*ABCD* = 353.4 + 643.718 + 111.479 (=111.479)	M1(DEP)		
	ABCD = 1108.6→ **1110**	A1		
	Method 4: $\Delta ABE + \Delta BED + \Delta BCD$		6	M1 M1 M1 M1 (DEP) M1 (DEP) A1
	Scheme: Δ*ABE* + Δ*BED* : M1(area formula for Δ*ABE*), M1(Δ*ABE* = Δ*BED*)			
	Δ*BCD*: M1(full method for ∠*DBC*), M1(area formula)			
	***ABCD*:** M1 (Adding areas), A1			
	$\Delta ABE + \Delta BED + \Delta BCD$			
	(*BE* = 20tan65 = 42.89)			
	$\lvert\Delta ABE\rvert = \frac{1}{2} \times 20 \times "42.89"$ (= 428.9)	M1		
	and $\lvert\Delta ABE\rvert = \lvert\Delta BED\rvert$ (Congruence)	M1		
	$\angle DBE = 25 \ \therefore \ \angle DBC = 70 - 25 = 45$	M1		
	$\lvert\Delta BCD\rvert = \frac{1}{2} \times 15 \times "47.324" \times \sin "45" \quad (= 250.97)$	M1(DEP)		
	ABCD = "428.9" + "428.9" + "250.97"	M1(DEP)		
	ABCD = 1108.77→**1110**			

Question		Scheme	Mark	Notes
11	(a)	$3x^4 - 11x^3 + 6x^2 + 9x - 6$ (Expanding, allow 1 slip) (OR $3\left(\frac{2}{3}\right)^4 + a\left(\frac{2}{3}\right)^3 + 6\left(\frac{2}{3}\right)^2 + 9\left(\frac{2}{3}\right) - 6 = 0$ (M1)) cc	2	M1 A1
	(b)	$\frac{dy}{dx} = 3x^2 - 6x$ (differentiating, one term correct) $"3x^2 - 6x" = 0$ $3x(x-2)$ (solving 2 term quadratic) (0, 3) and (2, -1) **NB:** Working must be seen	4	M1 M1 (DEP) M1 (DEP) A1
	(c)	(3), [Accept $-0.38, -0.375, -0.37, -\frac{3}{8}$], ($-1$), [Accept $-0.13, -0.125, -0.12, -\frac{1}{8}$], 1.11 [Accept $\frac{71}{64}$] **NB** : **(1)** Do **not** award respective A1 for (b) in (c). **(2)** 2dp answers required, penalise ONCE	3	B3 (-1eeoo)

Question	Scheme	Mark	Notes
(d)	Curve -1 mark for straight line segments each point missed each missed segment each point not plotted each point incorrectly plotted tramlines very poor curve **NB: (1)** Accuracy for both plotting and drawing is $\pm\frac{1}{2}ss$ = ±0.05 **(2)** Deduct errors starting with the last ePEN mark box	3	B3 (−1eeoo)
(e)	-0.88 (-0.91 to -0.85) 0.67 (Accept $\frac{2}{3}$), 1.35 (ie 1.32 to 1.38), 2.53 (ie 2.50 to 2.56)	4	B1 B1 B1 B1